What distinguishes Natural Language Generation (NLG) from classification tasks?
A. Selecting predefined outputs
B. Constructing novel sequences that preserve meaning and fluency
C. Assigning labels to tokens
D. Extracting key phrases

Which models are explicitly listed as examples for abstractive summarization?
A. BERT, RoBERTa, DeBERTa
B. BART, PEGASUS, T5
C. GPT, LLaMA, Falcon
D. CLIP, BLIP, Flamingo

What is the primary advantage of bottom-up summarization?
A. Eliminates need for neural models
B. Improves content control and reduces hallucination
C. Maximizes token-level likelihood
D. Removes extractive components

Which limitation of MLE training is explicitly mentioned?
A. High computational efficiency
B. Alignment with ROUGE metrics
C. Exposure bias and mismatch with evaluation metrics
D. Improved long-term coherence

What does REINFORCE optimize in RL-based summarization?
A. Token-level likelihood
B. Expected reward using Monte Carlo sampling
C. Cross-entropy loss only
D. Beam search objective

What is the role of the critic in Actor-Critic methods?
A. Generates summaries
B. Encodes input text
C. Estimates expected reward
D. Performs decoding

Why do Seq2Seq dialogue systems produce generic responses?
A. Due to reinforcement learning instability
B. Because likelihood maximization favors high-frequency responses
C. Because they lack attention mechanisms
D. Because they use beam search

What does Maximum Mutual Information (MMI) introduce?
A. Only $P(T|S)$
B. $P(S|T)$ and penalization of high-frequency responses
C. Only token-level likelihood
D. Beam size optimization

What is the purpose of n-gram blocking in dialogue systems?
A. Improve training speed
B. Prevent repeated phrases during decoding
C. Increase model size
D. Enhance attention weights

What is the role of the latent variable z_t in strategic-NLG separation?
A. Predicts surface text directly
B. Encodes token embeddings
C. Predicts future dialogue impact
D. Performs decoding

What happens during rollout-based decision making at test time?
A. Model selects first generated response
B. Model simulates future dialogue trajectories and selects highest expected reward

C. Model maximizes token probability only
D. Model disables reinforcement learning

What is the defining characteristic of Conversational Question Answering (CQA)?
A. Each question is processed independently
B. It maintains conversational context across multiple turns
C. It only uses retrieval-based methods
D. It ignores prior answers

Which challenge is explicitly associated with CQA?
A. Tokenization efficiency
B. Context tracking and resolving coreference
C. Beam search optimization
D. Vocabulary compression

What are the three primary modeling approaches used in CQA systems?
A. Classification, clustering, regression
B. Retrieval-based, generative, hybrid
C. CNN-based, RNN-based, Transformer-based
D. Supervised, unsupervised, reinforcement

Which architecture is used to store and retrieve conversation history in CQA?
A. CNN
B. Memory networks
C. Autoencoders
D. Graph neural networks

Which datasets are explicitly mentioned for conversational QA?
A. SQuAD, GLUE, SuperGLUE
B. ImageNet, COCO, VIST
C. CoQA, QuAC, DoQA
D. WikiText, BookCorpus, Common Crawl

What distinguishes conversational QA datasets from extractive QA benchmarks?
A. They only use retrieval
B. Answers are generated abstractively rather than copied verbatim
C. They do not use context
D. They only contain single-turn questions

What is a key requirement of storytelling in NLP?
A. Token-level probability maximization
B. Long-range coherence and causal structure
C. Only short sentence generation
D. Fixed vocabulary constraints

What is the effect of prompt structure in storytelling?
A. No impact on generation quality
B. Only affects tokenization
C. Significantly shapes narrative coherence and creativity
D. Reduces model size

What capability does multi-turn interactive prompting provide?
A. Removes context dependence
B. Enables collaborative storytelling with user-driven plot changes
C. Limits narrative diversity
D. Prevents long-range coherence

What is the purpose of Skip-Thought vectors?
A. Predict next token probability
B. Learn sentence representations by predicting surrounding sentences
C. Encode images into vectors
D. Perform syntactic parsing

What is the main advantage of event-based story generation?

A. Maximizes lexical overlap
B. Encourages causal progression and reduces repetitive surface-level generation
C. Eliminates need for neural models
D. Uses only word-level modeling

What does autoregressive decomposition reduce sequence modeling to?
A. Sequence classification
B. Next-token prediction
C. Token clustering
D. Sequence alignment

What is minimized during training of language models?
A. Perplexity directly
B. Entropy
C. Cross-entropy loss
D. Beam width

What limitation do RNNs face in modeling long-range dependencies?
A. Overfitting
B. Vanishing gradients
C. Lack of parameters
D. Limited vocabulary

What mechanism enables Transformers to model long-range dependencies?
A. Recurrence
B. Convolution
C. Self-attention
D. Pooling

What causes exposure bias?
A. Training uses ground-truth tokens but inference uses model predictions
B. Model overfits to data
C. Beam search errors
D. Tokenization mismatch

Why is exact decoding intractable?
A. Lack of training data
B. Memory limitations only
C. Numerical instability
D. Exponential growth of the search space

What is the key limitation of greedy decoding?
A. Locally optimal but globally suboptimal sequences
B. High computational cost
C. Requires large memory
D. Produces diverse outputs

What does beam search improve?
A. Training efficiency
B. Exploration by keeping multiple hypotheses
C. Tokenization quality
D. Vocabulary size

Why is length normalization applied?
A. Improve speed
B. Increase vocabulary
C. Mitigate bias toward shorter sequences
D. Reduce memory usage

What is the effect of temperature scaling?
A. Changes token order
B. Reshapes probability distribution sharpness
C. Eliminates randomness
D. Increases sequence length

What does perplexity measure?
A. Next-token prediction uncertainty
B. Sequence length
C. Beam size
D. Model depth

What is the goal of coreference resolution?
A. Token classification
B. Clustering mentions referring to the same entity
C. Sentence parsing
D. Language modeling

What is a mention in coreference resolution?
A. A token
B. A sentence
C. A noun phrase, pronoun, or proper noun
D. A document

What does MUC measure?
A. Links between mentions
B. Token accuracy
C. Sentence similarity
D. Embedding distance

What does CEAF optimize?
A. Token overlap
B. Sentence alignment
C. Feature similarity
D. Cluster alignment

What is the main limitation of mention-pair models?
A. Low accuracy
B. Lack of global consistency
C. High computation
D. Limited vocabulary

What do mention-ranking models improve?
A. Token prediction
B. Parsing
C. Antecedent selection consistency
D. Embedding size

What is the role of null antecedents?
A. Represent singleton mentions
B. Improve speed
C. Reduce parameters
D. Increase vocabulary

What is a key benefit of neural coreference models?
A. Rule-based reasoning
B. Learned contextual representations
C. Fixed features
D. No training required

What does discourse salience influence?
A. Tokenization
B. Embedding size
C. Pronoun resolution
D. Beam search

What challenge arises in cross-lingual coreference?
A. Tokenization only
B. Vocabulary size

C. Decoding
D. Morphological and syntactic variation

What is the main structural formulation of coreference?
A. Partitioning mentions into clusters
B. Sequence labeling
C. Classification
D. Regression

What shift did deep learning introduce?
A. Manual feature engineering
B. Automatic representation learning
C. Rule-based systems
D. Static embeddings

What is a limitation of single-task learning?
A. High accuracy
B. Fast training
C. Lack of knowledge sharing
D. Low computation

What is the purpose of pre-training?
A. Learn transferable representations
B. Reduce model size
C. Eliminate training
D. Improve tokenization

What is hard parameter sharing?
A. Separate models per task
B. Shared encoder with task-specific outputs
C. No sharing
D. Random sharing

What is negative transfer?
A. Improved performance
B. Faster training
C. Task interference due to conflicting gradients
D. Reduced parameters

What is the role of LoRA?
A. Increase parameters
B. Remove training
C. Improve tokenization
D. Reduce trainable parameters via low-rank updates

What does instruction tuning enable?
A. Single model handling multiple tasks via instructions
B. Separate models per task
C. Fixed outputs
D. No adaptation

What is catastrophic forgetting?
A. Improved performance
B. Loss of previous task knowledge
C. Faster inference
D. Reduced memory

What does zero-shot learning rely on?
A. Retraining
B. Labels
C. Pre-trained representations
D. Feature engineering

What is scalarization?

A. Removing objectives
B. Increasing parameters
C. Separate training
D. Combining multiple losses into a weighted sum

What defines Pareto optimality?
A. No task improves without harming another
B. Maximum accuracy
C. Minimum loss
D. Single objective optimization

What does semantic composition study?
A. Token prediction
B. How sentence meaning is built from parts
C. Parsing only
D. Classification

What property allows infinite embedding in language?
A. Attention
B. Tokenization
C. Recursion
D. Optimization

What does PP attachment ambiguity illustrate?
A. Tokenization issues
B. Training difficulty
C. Vocabulary mismatch
D. Structural ambiguity affecting meaning

What is a limitation of bag-of-words models?
A. Lack of role and structure encoding
B. High computation
C. Overfitting
D. Large memory

What does lambda calculus represent?
A. Word embeddings
B. Tokenization
C. Function-based semantic composition
D. Parsing trees

What is the goal of hybrid models?
A. Reduce parameters
B. Combine neural and symbolic approaches
C. Eliminate structure
D. Improve speed

What does SCAN evaluate?
A. Parsing accuracy
B. Embedding quality
C. Token prediction
D. Compositional generalization

How do Transformers compose meaning?
A. Tree recursion
B. Rule-based parsing
C. Self-attention
D. Bag-of-words

What does inductive bias influence?
A. Generalization behavior
B. Tokenization
C. Vocabulary
D. Speed

What distinguishes compositionality from memorization?
A. Dataset size
B. Ability to generalize to unseen combinations
C. Training time
D. Model size

What does scope ambiguity involve?
A. Token overlap
B. Sentence length
C. Parsing speed
D. Multiple interpretations due to quantifier scope

What defines autoregressive models?
A. Predict next token from prior context
B. Mask tokens
C. Encode full sequence
D. Retrieve documents

What is the objective of masked language modeling?
A. Predict next token
B. Reconstruct masked tokens
C. Retrieve text
D. Translate sequences

What characterizes Seq2Seq models?
A. Classification
B. Retrieval
C. Encode input and decode output
D. Masking

What is the purpose of retrieval augmentation?
A. Reduce parameters
B. Increase training speed
C. Improve tokenization
D. Incorporate external knowledge

What does scaling primarily improve?
A. Tokenization
B. Model performance
C. Vocabulary
D. Parsing

What is RLHF used for?
A. Token prediction
B. Retrieval
C. Aligning outputs with human preferences
D. Parsing

What is the role of dense retrievers in RAG?
A. Retrieve relevant documents
B. Generate text
C. Train model
D. Encode tokens

What challenge do multimodal models face?
A. Tokenization
B. Scaling
C. Decoding
D. Cross-modal alignment

What is quantization used for?
A. Increase parameters
B. Reduce memory and speed up inference

C. Improve accuracy
D. Train models

What is FlashAttention designed to address?
A. Tokenization
B. Vocabulary
C. Attention memory and compute efficiency
D. Training data

What is the advantage of MoE models?
A. Simplicity
B. Small size
C. Deterministic outputs
D. Large capacity with sparse computation

What distinguishes human cognition from machine learning systems?
A. Speed
B. Adaptability and abstraction
C. Data size
D. Tokenization

What is few-shot learning?
A. Large dataset training
B. Reinforcement learning
C. Learning from very few examples
D. Supervised training

What is OOD robustness?
A. Performance under distribution shifts
B. Training accuracy
C. Token prediction
D. Model size

What is a limitation of LLM explanations?
A. Too short
B. Too fast
C. Too large
D. May not reflect internal reasoning

What characterizes formal reasoning?
A. Heuristics
B. Strict logical rules
C. Probabilities
D. Tokens

What does Chain-of-Thought prompting do?
A. Generates intermediate reasoning steps
B. Changes model weights
C. Reduces parameters
D. Removes context

What is the main debate about LLM reasoning?
A. Speed vs size
B. Data vs compute
C. Reasoning vs statistical pattern completion
D. Tokens vs embeddings

What do reasoning benchmarks evaluate?
A. Tokenization
B. Speed
C. Vocabulary
D. Multi-step inference and consistency

What is a key limitation of LLM reasoning?

A. Low speed
B. Hallucination and inconsistency
C. Small models
D. Lack of tokens

What does generalization require?
A. Robustness beyond training distribution
B. More parameters
C. Faster training
D. Larger vocabulary

What does perplexity measure?
A. Reasoning ability
B. Model size
C. Prediction uncertainty
D. Data size

What is algorithmic bias?
A. Model speed
B. Bias learned from data patterns
C. Tokenization error
D. Hardware issue

What is a key privacy concern in NLP?
A. Speed
B. Memory size
C. Memorization of sensitive data
D. Tokenization

What is dual-use technology?
A. Beneficial and malicious applications
B. Only safe systems
C. Only harmful systems
D. Only academic systems

What does confidentiality refer to in the CIA triad?
A. Speed
B. Accuracy
C. Availability
D. Protection of sensitive data

What is representation bias?
A. Model architecture issue
B. Training error
C. Underrepresentation of groups in data
D. Tokenization

What is annotation bias?
A. Hardware limitation
B. Subjective labeling decisions
C. Model size
D. Vocabulary

What is counterfactual evaluation used for?
A. Training
B. Optimization
C. Tokenization
D. Bias detection via input variation

What is demographic parity?
A. Equal prediction rates across groups
B. Equal data size
C. Equal training
D. Equal tokens

What does equalized odds enforce?
A. Equal predictions
B. Equal data
C. Equal error rates across groups
D. Equal speed

What is a key limitation of fairness definitions?
A. Too simple
B. They may conflict with each other
C. Too fast
D. Too accurate

What supports accountability in NLP systems?
A. Larger models
B. Faster training
C. More data
D. Model cards and documentation

Natural Language Generation (NLG) systems synthesize new text rather than selecting predefined outputs.
Abstractive summarization generates new paraphrased summaries using neural models such as BART, PEGASUS, and T5.
Extractive summarization selects key sentences or phrases directly from the source text.
Bottom-up summarization first selects salient content and then rewrites it into a coherent summary.
Reinforcement Learning (RL) for summarization optimizes models by maximizing long-term rewards rather than token-level likelihood.
Maximum Likelihood Estimation (MLE) training aligns directly with evaluation metrics such as ROUGE.
Exposure bias occurs because during training the model observes gold-standard summaries but at inference time generates tokens autoregressively.
Self-Critical Sequence Training (SCST) uses the model’s greedy-decoded summary as a baseline to reduce gradient variance.
Actor-Critic methods include an actor that generates summaries and a critic that estimates expected reward.
Higher ROUGE scores always correspond to improved human readability.
Maximum Mutual Information (MMI) modifies the objective by incorporating P(S | T) and penalizing high-frequency responses.
Conversational Question Answering (CQA) treats each question independently without maintaining conversational context.
CQA requires understanding previous questions, prior answers, and evolving discourse state.
Retrieval-based systems dynamically generate responses using language models such as GPT or BART.
Hybrid CQA approaches combine retrieval and generation, as in Retrieval-Augmented Generation (RAG).
Memory networks are used to store and retrieve conversation history in CQA systems.
Conversational QA datasets such as CoQA and QuAC involve multi-turn question answering over context.
Storytelling in NLP requires modeling plot progression, character development, and stylistic consistency.
Prompt design has no significant effect on storytelling quality.
Multi-turn interactive prompting allows users to introduce plot twists that the model continues dynamically.
Skip-Thought vectors learn sentence representations by predicting neighboring sentences in a corpus.
Event-based story generation represents narratives as sequences of structured events rather than raw token sequences.
Language modeling reduces sequence probability estimation to autoregressive next-token prediction.
Conditional language modeling generates a target sequence independent of input context.
n-gram models approximate conditional probabilities using a fixed context window.
Transformer-based models rely on recurrence rather than self-attention.
Teacher forcing supplies the ground-truth token during training when predicting the next token.
Exposure bias arises because training and inference use the same distribution.
Decoding is a search problem over an exponentially large hypothesis space.
Greedy decoding guarantees globally optimal sequences.
Beam search retains multiple high-scoring hypotheses during decoding.
Top-k sampling eliminates all stochasticity in decoding.
Perplexity measures token-level prediction accuracy but does not guarantee high-quality generation.
Coreference resolution is the task of clustering mentions that refer to the same real-world entity.
Anaphora includes all forms of coreference without restriction.
Coreference resolution is fundamentally a clustering problem over textual mentions.
MUC evaluates clustering using mention-level overlap.

B^3 computes precision and recall at the mention level.
CEAF aligns predicted and gold clusters using optimal one-to-one matching.
Mention-pair models guarantee global consistency across clusters.
Mention-ranking models select the best antecedent for each mention.
Coreference defines an equivalence relation that must satisfy reflexivity, symmetry, and transitivity.
Self-attention enables modeling of long-distance dependencies across documents.
Coreference resolution depends only on lexical similarity and not on world knowledge.
Deep learning enables automatic feature learning from raw data.
Single-task learning promotes knowledge sharing across tasks.
Pre-training learns general-purpose representations on large datasets.
Multi-task learning integrates multiple objectives into a shared parameter space.
Hard parameter sharing eliminates negative transfer between tasks.
Soft parameter sharing maintains separate models while regularizing their similarity.
Negative transfer occurs when gradients from tasks conflict.
Fine-tuning always preserves previously learned knowledge.
LoRA reduces trainable parameters by factorizing weight updates into low-rank matrices.
Instruction tuning requires separate output heads for each task.
Multi-task learning can be formulated as a multi-objective optimization problem.
Semantic composition derives sentence meaning from the structured combination of its parts.
Meaning arises solely from individual words without considering structure or context.
Recursion allows infinite embedding of phrases within phrases.
Center embedding is cognitively easy for humans to process.
Parsing ambiguity directly affects semantic interpretation.
Distributional models handle logical operators such as quantifiers and negation effectively.
Tree-based neural models explicitly encode hierarchical structure.
Transformers enforce explicit tree structures during semantic composition.
Formal semantics models meaning as function application aligned with syntax.
Hybrid neuro-symbolic models combine structured logic with neural representations.
Compositional generalization requires systematic recombination beyond memorized patterns.
Autoregressive models predict the next token given prior context.
Autoencoding models use masked language modeling with bidirectional context.
Sequence-to-sequence models encode input and decode output sequences.
Retrieval-augmented models rely only on parametric knowledge stored in model weights.
Multimodal models integrate text with image, audio, or video representations.
Instruction-aligned models use supervised fine-tuning and preference optimization.
Causal language modeling uses bidirectional context during training.
Scaling laws show predictable performance improvements with increased model size and data.
Retrieval-augmented generation reduces hallucination by conditioning on external documents.
Self-attention scales linearly with sequence length.
Mixture-of-Experts activates all parameters for every token.
Human intelligence exhibits strong few-shot learning and adaptability to novel tasks.
Machine learning systems by design possess structured reasoning mechanisms.
Few-shot generalization in machines remains an open research challenge.
Language models are robust to distribution shifts and minor input perturbations.
Generated explanations from LLMs always reflect the true internal reasoning process.
Formal reasoning guarantees validity through strict inference rules.
LLMs primarily operate within informal reasoning regimes.
Chain-of-Thought prompting modifies model parameters during inference.

Chain-of-Thought prompting can improve performance on multi-step reasoning tasks.
Perplexity directly measures reasoning ability in language models.
Emergent abilities refer to nonlinear improvements in performance with scale.
Ethical risks in NLP arise from data, model architecture, deployment context, and societal factors.
Algorithmic bias can originate from representation, historical, and measurement bias.
Publicly available data always implies ethical permissibility for model training.
Generative models can be used to produce misinformation and automated propaganda.
NLP systems are inherently transparent and easy to audit.
The CIA triad consists of confidentiality, integrity, and availability.
Bias in NLP systems is limited to dataset imbalance only.
Cognitive biases in human language can be inherited by NLP models.
Bias can arise during data collection, annotation, model training, and deployment.
Fairness definitions such as demographic parity and equalized odds are always compatible.
Bias mitigation is a continuous process across the NLP lifecycle.
This has been a difficult course.